

$$\begin{aligned}
& \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{3,0} [m_2] + \frac{1}{3} \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{4,-1} [m_2] - \frac{16}{45} \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{5,-2} [m_2] + \\
& \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^2 (9 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} + 4 \sum_{\mathbf{p}} \mathcal{G}_2^2) \mathcal{L} \mathcal{F}_{3,0} [m_1^{\text{p}}] - \frac{1}{24} \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^2 (6 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} + 5 \sum_{\mathbf{p}} \mathcal{G}_2^2) \mathcal{L} \mathcal{F}_{4,-1} [m_1^{\text{p}}] + \\
& \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{5,-2} [m_1^{\text{p}}] + \frac{1}{12} \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^2 (9 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} - 9 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} + 4 \sum_{\mathbf{p}} \mathcal{G}_2^2) \mathcal{L} \mathcal{F}_{3,0} [m_q^{\text{p}}] - \\
& \mathcal{G}_2^2 (6 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} - 6 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} + 5 \sum_{\mathbf{p}} \mathcal{G}_2^2) \mathcal{L} \mathcal{F}_{4,-1} [m_q^{\text{p}}] + \frac{4}{15} \sum_{\mathbf{p}} \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{5,-2} [m_q^{\text{p}}] + \\
& \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{3,0} [\tilde{\mu}] + \frac{1}{6} \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{4,-1} [\tilde{\mu}] - \frac{8}{45} \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{5,-2} [\tilde{\mu}] - m_1 \tilde{\mu} \mathcal{G}_1^4 \mathcal{L} \mathcal{F}_{2,2,-1} [m_1, \tilde{\mu}] - \\
& \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{2,1,0} [m_2, \tilde{\mu}] - \frac{1}{12} \mathcal{G}_2^4 (13 \mathcal{C}_{\phi 1 \phi 2} + 36 m_2 \tilde{\mu}) \mathcal{L} \mathcal{F}_{2,2,-1} [m_2, \tilde{\mu}] + \\
& \mathcal{G}_2^4 (-3 m_2 \mathcal{C}_{\phi 1 \phi 2} + 2 \tilde{\mu} (\mathcal{C}_{\phi 1^2} + 3 \mathcal{C}_{\phi 2^2})) \mathcal{L} \mathcal{F}_{2,2,0} [m_2, \tilde{\mu}] + \\
& \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{3,1,-1} [m_2, \tilde{\mu}] + \frac{2}{3} \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^4 \mathcal{L} \mathcal{F}_{3,2,-2} [m_2, \tilde{\mu}] - \\
& \mathcal{G}_2^4 (\mathcal{C}_{\phi 1^2} + 3 \mathcal{C}_{\phi 2^2}) \mathcal{L} \mathcal{F}_{3,2,-1} [m_2, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} \mathcal{G}_1^2 \overline{y}_d^{\text{pr}} a_d^{\text{pr}} \mathcal{L} \mathcal{F}_{2,1,0} [m_d^{\text{r}}, m_q^{\text{p}}] + \\
& (\mathcal{C}_{\phi 1 \phi 2} \overline{a}_d^{\text{pr}} a_d^{\text{pr}} + \tilde{\mu} \overline{y}_d^{\text{pr}} (a_d^{\text{pr}} (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) + \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_d^{\text{pr}})) \mathcal{L} \mathcal{F}_{2,2,0} [m_d^{\text{r}}, m_q^{\text{p}}] + \\
& \overline{y}_e^{\text{pr}} a_e^{\text{pr}} \mathcal{L} \mathcal{F}_{2,1,0} [m_e^{\text{r}}, m_l^{\text{p}}] + \\
& (\mathcal{C}_{\phi 1 \phi 2} \overline{a}_e^{\text{pr}} a_e^{\text{pr}} + \tilde{\mu} \overline{y}_e^{\text{pr}} (a_e^{\text{pr}} (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) + \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_e^{\text{pr}})) \mathcal{L} \mathcal{F}_{2,2,0} [m_e^{\text{r}}, m_l^{\text{p}}] - \\
& \overline{y}_e^{\text{pr}} a_e^{\text{pr}} \mathcal{L} \mathcal{F}_{2,1,0} [m_l^{\text{p}}, m_e^{\text{r}}] + \frac{1}{4} \tilde{\mu} \overline{y}_e^{\text{pr}} a_e^{\text{pr}} (\mathcal{G}_1^2 + \mathcal{G}_2^2) \mathcal{L} \mathcal{F}_{3,1,-1} [m_l^{\text{p}}, m_e^{\text{r}}] + \\
& \mathcal{C}_{\phi 2} \overline{a}_e^{\text{pr}} a_e^{\text{pr}} (-\mathcal{G}_1^2 + \mathcal{G}_2^2) - 2 \tilde{\mu} \mathcal{G}_1^2 \overline{y}_e^{\text{pr}} (a_e^{\text{pr}} (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) + \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_e^{\text{pr}})) \mathcal{L} \mathcal{F}_{3,1,0} [m_l^{\text{p}}, m_e^{\text{r}}] - \\
& (\mathcal{C}_{\phi 1 \phi 2} \overline{a}_e^{\text{pr}} a_e^{\text{pr}} + \tilde{\mu} \overline{y}_e^{\text{pr}} (a_e^{\text{pr}} (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) + \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_e^{\text{pr}})) \mathcal{L} \mathcal{F}_{3,2,-1} [m_l^{\text{p}}, m_e^{\text{r}}] + \\
& \mathcal{C}_{\phi 2} \overline{a}_e^{\text{pr}} a_e^{\text{pr}} (3 \mathcal{G}_1^2 + \mathcal{G}_2^2) + \tilde{\mu} \overline{y}_e^{\text{pr}} (a_e^{\text{pr}} (9 \mathcal{G}_1^2 (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) + \mathcal{G}_2^2 (5 \mathcal{C}_{\phi 1^2} + 9 \mathcal{C}_{\phi 2^2})) + \\
& \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_e^{\text{pr}} (9 \mathcal{G}_1^2 + 5 \mathcal{G}_2^2))) \mathcal{L} \mathcal{F}_{4,1,-1} [m_l^{\text{p}}, m_e^{\text{r}}] + \frac{1}{6} (-3 \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} + \\
& \overline{y}_e^{\text{pr}} (a_e^{\text{pr}} (-3 \mathcal{G}_1^2 (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) - 2 \mathcal{G}_2^2 (2 \mathcal{C}_{\phi 1^2} + 3 \mathcal{C}_{\phi 2^2})) - \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_e^{\text{pr}} (3 \mathcal{G}_1^2 + 4 \mathcal{G}_2^2))) \\
& , -2 [m_l^{\text{p}}, m_e^{\text{r}}] - \frac{1}{2} \mathcal{C}_{\phi 1 \phi 2} \overline{y}_e^{\text{pr}} \overline{y}_e^{\text{st}} y_e^{\text{pt}} y_e^{\text{sr}} \mathcal{L} \mathcal{F}_{2,1,0} [m_l^{\text{p}}, m_l^{\text{s}}] + \\
& \overline{y}_e^{\text{pr}} \overline{y}_e^{\text{st}} y_e^{\text{pt}} y_e^{\text{sr}} \mathcal{L} \mathcal{F}_{3,1,-1} [m_l^{\text{p}}, m_l^{\text{s}}] - \frac{1}{2} \tilde{\mu} \mathcal{G}_1^2 \overline{y}_d^{\text{pr}} a_d^{\text{pr}} \mathcal{L} \mathcal{F}_{2,1,0} [m_q^{\text{p}}, m_d^{\text{r}}] + \\
& \overline{y}_d^{\text{pr}} a_d^{\text{pr}} (\mathcal{G}_1^2 + \mathcal{G}_2^2) \mathcal{L} \mathcal{F}_{3,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}] + \frac{1}{4} \\
& - 2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} (\mathcal{G}_1^2 + 3 \mathcal{G}_2^2) - 2 \tilde{\mu} \mathcal{G}_1^2 \overline{y}_d^{\text{pr}} (a_d^{\text{pr}} (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) + \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_d^{\text{pr}})) \mathcal{L} \mathcal{F}_{3,1,0} [m_q^{\text{p}}, m_d^{\text{r}}] - \\
& (\mathcal{C}_{\phi 1 \phi 2} \overline{a}_d^{\text{pr}} a_d^{\text{pr}} + \tilde{\mu} \overline{y}_d^{\text{pr}} (a_d^{\text{pr}} (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) + \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_d^{\text{pr}})) \mathcal{L} \mathcal{F}_{3,2,-1} [m_q^{\text{p}}, m_d^{\text{r}}] + \\
& \mathcal{C}_{\phi 2} \overline{a}_d^{\text{pr}} a_d^{\text{pr}} (-\mathcal{G}_1^2 + \mathcal{G}_2^2) + \\
& \overline{y}_d^{\text{pr}} (a_d^{\text{pr}} (5 \mathcal{G}_1^2 (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) + \mathcal{G}_2^2 (5 \mathcal{C}_{\phi 1^2} + 9 \mathcal{C}_{\phi 2^2})) + 5 \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_d^{\text{pr}} (\mathcal{G}_1^2 + \mathcal{G}_2^2))) \\
& , -1 [m_q^{\text{p}}, m_d^{\text{r}}] + \frac{1}{2} (-3 \mathcal{C}_{\phi 1 \phi 2} \mathcal{G}_2^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} + \\
& \overline{y}_d^{\text{pr}} (a_d^{\text{pr}} (-3 \mathcal{G}_1^2 (\mathcal{C}_{\phi 1^2} + \mathcal{C}_{\phi 2^2}) - 2 \mathcal{G}_2^2 (2 \mathcal{C}_{\phi 1^2} + 3 \mathcal{C}_{\phi 2^2})) - \mathcal{C}_{\phi 1 \phi 2} \tilde{\mu} y_d^{\text{pr}} (3 \mathcal{G}_1^2 + 4 \mathcal{G}_2^2))) \\
& , -2 [m_q^{\text{p}}, m_d^{\text{r}}] + \frac{3}{4} \mathcal{C}_{\phi 1 \phi 2} \overline{y}_d^{\text{pr}} y_d^{\text{sr}} (-2 \overline{y}_d^{\text{st}} y_d^{\text{pt}} + \overline{y}_u^{\text{st}} y_u^{\text{pt}}) \mathcal{L} \mathcal{F}_{2,1$$